

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and optimized **Python** ETL processes to transform diverse healthcare datasets into clean, structured formats for machine learning model training, processing approximately **1.5 TB** of raw data weekly.
- Designed and implemented automated CI/CD pipelines for machine learning workflows using **MLflow** and **Git**, ensuring robust testing, version control, and streamlined model deployment cycles.
- Deployed machine learning models as scalable **APIs** and microservices using **Docker** and **Kubernetes**, meeting stringent clinical performance and latency requirements for real-time inference.
- Implemented production monitoring solutions for AI services, tracking key metrics like model performance, data drift, and system health using **Prometheus** and **Grafana**.
- Wrote clean, maintainable, and well-documented **Python** code, actively participating in code reviews to uphold system reliability and adherence to engineering best practices.
- Collaborated with backend engineering teams to integrate machine learning outputs into core healthcare applications, ensuring seamless data flow and functional compatibility across systems.
- Managed and maintained feature stores and data layers using **PostgreSQL**, ensuring consistency between training and production environments and reducing data discrepancy issues by **30%**.

SKILLS

Programming – Python, SQL, Java, Go

Cloud & Infrastructure – AWS, Docker, Kubernetes, GCP

ML Tools – MLflow, Scikit-learn, PyTorch, Kubeflow, Airflow

Data Tools – Pandas, Spark, dbt, PostgreSQL, FHIR, HL7

MLOps & Deployment – CI/CD, API Deployment, Microservices, Model Serving, Monitoring

Security & Compliance – HIPAA, HITRUST

PROJECTS

ML Pipeline Orchestration for Healthcare Data

- Engineered a comprehensive data pipeline using **Python** and **Pandas** to ingest and clean **FHIR** and **HL7** data from multiple sources, preparing datasets for model training.
- Implemented automated model training and evaluation workflows with **Scikit-learn**, achieving an **AUC of 0.88** on a simulated patient risk prediction task.
- Containerized the entire ML workflow using **Docker** and orchestrated deployments on a **Kubernetes** cluster, ensuring consistent and scalable execution.

Production Model Serving and Monitoring System

- Developed scalable **Python** microservices using **FastAPI** to serve trained ML models, handling approximately **500 requests per second** with p95 latency under **200ms**.
- Integrated real-time monitoring with **Prometheus** and **Grafana** to track model inference performance and detect data drift, enabling proactive alerts for **~15 critical metric deviations** per week.
- Managed model versioning and deployment strategies using **MLflow**, facilitating rollbacks and A/B testing of different model iterations in a production-like environment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)